

## Zonglin Di

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| CONTACT INFORMATION     | Tongji University, 1239 Siping Road<br>Yangpu District, Shanghai, P. R. China, 201800                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | (+86) 185-1669-7168<br>1452640@tongji.edu.cn |
| RESEARCH INTEREST       | Geographic knowledge discovery with deep learning, Automatic control with graph neural network (GNN), Weakly-supervised learning, Computer Vision                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                              |
| EDUCATION               | <b>Tongji University</b> , Shanghai, China<br><br>Bachelor of Engineering, <b>Control Science and Engineering</b> , Sept 2014 to July 2019 <ul style="list-style-type: none"><li>• GPA: 4.27/5.0 (87.7/100)</li><li>• Related Courses: Higher Mathematics, Linear Algebra, Possibility Theory &amp; Statistics, Combinatorial Mathematics, Pattern Recognition, Object-Oriented Programming, Database Systems, Operating System, Compiler Principle, Digital Signal Processing, Digital Image Processing, Analog/Digital Circuits, Signal &amp; Systems</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                              |
| RESEARCH EXPERIENCE     | <b>Research Assistant of Deep Learning Laboratory</b> June 2018 to Present <ul style="list-style-type: none"><li>• Perform large-scale roadmap extraction training on 40+ GPUs distributed training and rough labeling to study the limitation of weakly-supervised learning in semantic segmentation</li><li>• Improve the performance of road extraction using deep learning and GPS data, allowing 5% increase in accuracy and 40% boost in generalization ability for new area prediction</li><li>• Advisor: Prof. Yin Wang, PhD of University of Michigan</li></ul> <b>Research Intern in Intel Asia &amp; Pacific R &amp; D</b> Oct 2017 to Mar 2018 <ul style="list-style-type: none"><li>• Improved the performance of existing semantic segmentation algorithms for removing human portrait from an image and patching the vacant area with variational auto encoders (VAE)</li><li>• Directed by Mr. Jiajie Yao</li></ul> <b>Research Assistant of State Lab of Embedded System &amp; Service Computing, Ministry of Education</b> Dec 2014 to Sept 2019 <ul style="list-style-type: none"><li>• Developed a novel method to facilitate the detection of Internet addiction disorder (IAD) so as to reduce the subjectivity compared to the traditional methods</li><li>• Completed a systematic test, showing the new model's capability to identify optimal feature combination in four questionnaires, with the accuracy of IAD labeling reaching 96.2%</li><li>• Advisor: Prof. Asoke K. Nandi, <i>IEEE Fellow</i></li></ul> <b>Research Assistant of State Key Lab of Management &amp; Control for Complex Systems, Ministry of Education</b> Sept 2016 to Sept 2019 <ul style="list-style-type: none"><li>• Developed a hybrid method for binary and multi-class imbalance classification based on blending support vector machine (SVM) and clustering algorithms and using density peak as the metric</li><li>• Incorporated data distribution and data imbalance simultaneously for improving the imbalance classification of SVM</li><li>• Supervisors: Prof. Mengchu Zhou, <i>IEEE Fellow</i> &amp; Prof. Qi Kang, <i>IEEE Senior Member</i></li></ul> |                                              |
| CONFERENCE PUBLICATIONS | 1. <b>Zonglin Di</b> , Qi Kang, Daogang Peng and Mengchu Zhou, "Density Peak based Pre-clustering Support Vector Machine for Multi-class Imbalanced Classification" <i>IEEE Conference on System Man and Cybernetics (SMC)</i> , pp. 27-32, 2019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                              |

2. Tao Sun, **Zonglin Di**, Pengyu Che, Chun Liu and Yin Wang, “Leveraging Crowdsourced GPS Data for Road Extraction from Aerial Imagery”  
*IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 7509-7518, 2019
3. Tao Sun, **Zonglin Di** and Yin Wang, “Combining Satellite Imagery and GPS Data for Road Extraction”  
*2nd ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery (GeoAI)*, pp. 29-32, 2018
4. **Zonglin Di**, Siya Yao, Qi Kang and Mengchu Zhou, “AP-SVM: An Adaptive Clustering-based Support Vector Machine Algorithm for Imbalance Classification”  
*IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 681-686, 2018
5. **Zonglin Di** and Qi Kang, “ADPclust-SVM: An Adaptive Clustering-based Support Vector Machine Algorithm for Imbalance Classification” (Abstract Accepted)  
*IEEE International Conference on Networking, Sensing and Control (ICNSC)*, 2018
6. **Zonglin Di**, Xiaoliang Gong, Jingyu Shi, Hosameldin O. A. Ahmed and Asoke K. Nandi, “Detection of Internet Addiction Disorder Based on Personality Questionnaires of Chinese College Students and SVMs”  
*10th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (ICSP-BMEI)*, pp. 1-6, 2017
7. Xiaoliang Gong, Bozhong Long, Kun Fang, **Zonglin Di**, Yichu Hou and Lei Cao, “A Prediction Based on Clustering and Personality Questionnaire Data for IGD Risk: A Preliminary Work”  
*12th International Conference on Neural Computation, Fuzzy System and Knowledge Discovery (ICNC-FSKD)*, pp. 1699-1703, 2016

JOURNAL  
PUBLICATIONS

1. **Zonglin Di**, Xiaoliang Gong, Jingyu Shi, Hosameldin O. A. Ahmed and Asoke K. Nandi, “Internet Addiction Disorder Detection of Chinese College Students Using Several Personality Questionnaire Data and Support Vector Machine”  
*Addictive Behavior Reports*, Elsevier, 2019

TEACHING  
ASSISTANT

**Python Programming (10064701)** Sept 2018 to Nov 2018  
**Deep Learning (10068501)** Mar 2019 to June 2019  
 Department of Computer Science and Technology, Tongji University  
 Supervised by Prof. Yin Wang  
**Open Source Hardware and Programming (55010501)** Sept 2018 to Jan 2019  
 School of Design and Innovation, Tongji University  
 Supervised by Prof. Xiaohua Sun

INTERN  
EXPERIENCES

**Full-Stack Software Intern in National Instruments, Shanghai** April 2018 to April 2019  
 Established framework of Auto Build System in the Academic Group using React.js, Flask, and JQuery  
 Implemented frontend and backend within one-month timeframe independently  
 Directed by Mr. Dasheng Xu  
**Software Intern in National Instruments, Shanghai** July 2017 to Oct 2017  
 Diadem data plug-in from German and Japanese customers using C++  
 Directed by Ms. Mavis Jin  
**Software Intern in Microsoft, Shanghai** July 2016 to Sept 2016

Built ASP.NET WebSite and the backend engine  
 Directed by Mr. Tarry Wang and Alex Zhang  
**Technical Marketing Intern in Jikexueyuan, Beijing**      July 2015 to Sept 2015  
 Solved the Android issues in the community

SELECTED  
AWARDS

- The 2nd Price, 30th National “Challenge Cup” Academic Science and Technology Competition      Nov 2019
- Special Award, 30th “Challenge Cup” Academic Science and Technology Competition Shanghai      June 2019
- The Honorable Prize, 2019 Mathematical Contest in Modeling (MCM)      Feb 2019
- The 2nd Price, Shanghai Collegiate Maker Competition      Jan 2018
- The 3rd Prize, China Undergraduate Mathematical Contest in Modeling (Shanghai Division)      Nov 2017
- Social Practice Scholarship, Tongji University      Sept 2017
- The 3rd Prize, The 5th “Taidi Cup” Data Mining Contest      Sept 2017
- The 3rd Prize, The 15th Digilent Design Contest      July 2017
- The 3rd Prize, The 1st “Huichuangqingchun” Culture & Creativity Project      Mar 2016

SELECTED  
PRESENTATIONS

- SMC 2019, Bari, Italy      Oct 2019
- CVPR 2019, Long Beach, USA      June 2019
- GeoAI 2018, Seattle, USA      Nov 2018
- SMC 2018, Miyazaki, Japan      Oct 2018
- ICNSC 2018, Zhuhai, China      Mar 2018
- ICSP-BMEI 2017, Shanghai, China      Sept 2017
- ICNC-FSKD 2016, Changsha, China      Aug 2016
- Excellent Undergraduate, Tongji University      Sept 2015, 2016, 2017

SELECTED SOCIAL EXPERIENCE

**IEEE CVPR 2019 Student Volunteer**      June 15 to June 21, 2019  
 Helped guide the way to the different session for the participants  
 Helped organize the welcome party

**Virtual Instruments Club**      Sept 2015 to Sept 2018  
**Member of Technical Department** (Sept 2015 to Jan 2016) → **Minister of Technical Department** (Jan 2016 to Jan 2017) → **Vice President** (Jan 2017 to Sept 2017) → **President** (Sept 2017 to Sept 2018)  
 Collaborated with National Instruments (Great China Area) in organizing multiple on-campus events at Tongji University  
 Delivered 20+ presentations to 200+ students

**Google Developer Group**      Dec 2016 to Sept 2018  
 Invited to Google Developer Days in 2016 and 2017  
 Organized the Google DevFes 2017  
 Organized the Google TensorFlow Technical Salon on the application of deep learning in computer vision and natural language processing

**Unity Unite, Shanghai and Beijing**      May 2017 and May 2018  
 Keynote and Tech Conference Track Support

REVIEWER EXPERIENCE

- The 2th International Conference on Mechanical, Electric and Industrial Engineering
- ACM SIGSPATIAL 2019 (External)